

Econ 771 hw 3

2. First we need to show the unbiasedness of W_n , in which we have $E(W_n) = \mu$:

$$E(W_n) = \frac{2}{n(n+1)} \sum_{i=1}^n i E(X_i) \text{, since } X_i \text{ are i.i.d with } E(X_i) = \mu, \forall i,$$

$$E(W_n) = \frac{2\mu}{n(n+1)} \underbrace{\sum_{i=1}^n i}_{:= \frac{n(n+1)}{2}} = \frac{2\mu n(n+1)}{2n(n+1)} = \mu.$$

$$\text{Then, because } X_i \text{ are i.i.d, } \text{Var}(W_n) = \sum_{i=1}^n \text{Var}\left(\frac{2i}{n(n+1)} X_i\right)$$

(if X, Y indep. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$) indep. from X , can be viewed as a constant

$$\dots = \sum_{i=1}^n \frac{4i^2}{n^2(n+1)^2} \text{Var}(X_i) = \frac{4}{n^2(n+1)^2} \cdot \sum_{i=1}^n i^2 \text{Var}(X_i)$$

(as $\text{Var}(aX) = a^2 \text{Var}(X)$ for some constant a)

$$= \frac{4}{n^2(n+1)^2} \cdot \delta^2 \cdot \underbrace{\sum_{i=1}^n i^2}_{:= \frac{n(n+1)(2n+1)}{6}} = \frac{4n(n+1)(2n+1)\delta^2}{6n^2(n+1)^2} = \frac{4(2n+1)}{6n(n+1)} \cdot \delta^2.$$

as $n \rightarrow \infty$, we have $\text{Var}(W_n) \rightarrow 0$

By Chebyshev's inequality that $\forall \epsilon > 0, \mathbb{P}(|W_n - E(W_n)| > \epsilon) \leq \frac{\text{Var}(W_n)}{\epsilon^2}$

give that we have $n \geq 1$, hence W_n always have finite variances.

Hence, when $n \rightarrow \infty, \mathbb{P}(|W_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(W_n)}{\epsilon^2} \rightarrow 0$

$\dots \mathbb{P}(|W_n - \mu| \geq \epsilon) \rightarrow 0$ when $n \rightarrow \infty, \forall \epsilon > 0$. Hence, $W_n \xrightarrow{P} \mu$.

3. Since we have $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$, we can rewrite that $\hat{\theta}_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(\hat{\theta}_n - \theta)$

we can show that $\sqrt{n}(\hat{\theta}_n - \theta) = O_p(1)$ (bounded in probability),

because $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$

Lemma: convergence in distribution implies $O_p(1)$.

if $Z_n \xrightarrow{d} Z$ for some finite R.V. Z , then $Z_n = O_p(1)$.

$\forall \epsilon > 0$, we can choose M s.t. $P(|Z| > M) < \frac{\epsilon}{2}$

as $Z_n \xrightarrow{d} Z$, the tail prob. converges, hence $\exists N \in \mathbb{N}$ s.t. $\forall n > N, \mathbb{P}(|Z_n| > M) \leq \mathbb{P}(|Z| > M) + \frac{\epsilon}{2} < \epsilon \dots \forall n > N, \mathbb{P}(|Z_n| > M) < \epsilon \dots Z_n = O_p(1)$

Then, as when $n \rightarrow \infty$, $\frac{1}{\sqrt{n}} \rightarrow 0$, hence $\frac{1}{\sqrt{n}} = o(1)$ goes to zero. (deterministic)

Furthermore, we then have $\hat{\theta}_n - \theta = o_p(1) \dots \hat{\theta}_n \xrightarrow{P} \theta$.

$\dots$ goes to zero in probability

(this is because $\hat{\theta}_n - \theta = O_p(1) \cdot o(1)$, by Slutsky's theorem, if $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (A R.V. b constant), then $A_n B_n \xrightarrow{d} Ab$.)

$\dots$ as $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2)$, $\frac{1}{\sqrt{n}} \xrightarrow{P} 0$, then $\hat{\theta}_n - \theta \xrightarrow{d} 0 \dots$ as 0 constant, $\hat{\theta}_n - \theta \xrightarrow{P} 0$, $\hat{\theta}_n \xrightarrow{P} \theta$.)

4. First, with finite sample $(X_t)_{t=1}^n$, we can construct a natural estimator

$\hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^n X_t^2$ The mean and variance of this estimator is

$$\mathbb{E}(\hat{\sigma}^2) = \frac{1}{n} \sum_{t=1}^n \mathbb{E}(X_t^2) = \frac{1}{n} \sum_{t=1}^n \sigma^2 = \sigma^2$$

σ^2 , as $\mathbb{E}(X_t) = 0$, $\text{Var}(X_t) = \mathbb{E}(X_t^2) - (\mathbb{E}(X_t))^2$

$$\text{Var}(\hat{\sigma}^2) = \mathbb{E}((\hat{\sigma}^2)^2) - (\mathbb{E}(\hat{\sigma}^2))^2 = \mathbb{E}(\hat{\sigma}^4) - \sigma^4$$

As $X_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$, we can have $X_t = \sigma Z_t$, in which $Z_t \stackrel{i.i.d.}{\sim} N(0, 1)$

$$\text{Hence, } \hat{\sigma}^2 = \frac{1}{n} \sum_{t=1}^n (\sigma Z_t)^2 = \frac{\sigma^2}{n} \sum_{t=1}^n Z_t^2 = \frac{\sigma^2}{n} Y$$

$:= Y$, and $Y \sim \chi_n^2$ by def. of χ^2 dist.

Then, $\mathbb{E}(\hat{\sigma}^4) = \mathbb{E}((\hat{\sigma}^2)^2) = \frac{\sigma^4}{n^2} \cdot \mathbb{E}(Y^2)$ By the property of χ^2 dist. we have

$$\mathbb{E}(Y) = n, \text{Var}(Y) = 2n, \text{hence } \mathbb{E}(Y^2) = n^2 + 2n \text{ Therefore } \mathbb{E}(\hat{\sigma}^4) = \sigma^4 \frac{n^2 + 2n}{n^2} = \sigma^4 \left(1 + \frac{2}{n}\right)$$

$$\text{Therefore, } \text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n}$$

This can be derived from the fact that $\text{Var}(Y) = \sum_{t=1}^n \text{Var}(Z_t^2)$

Since $Z_t \stackrel{i.i.d.}{\sim} N(0, 1)$

Give that MGF for a standard normal is $M_Z(a) = \mathbb{E}(e^{aZ}) = e^{\frac{a^2}{2}}$

and $\mathbb{E}(Z^k) = M_Z^{(k)}(0)$, as $\begin{cases} M_Z'(a) = a e^{\frac{a^2}{2}}; M_Z''(a) = (1+a^2) e^{\frac{a^2}{2}} \\ M_Z'''(a) = (3a+a^3) e^{\frac{a^2}{2}}, M_Z^{(4)}(a) = (3+6a^2+a^4) e^{\frac{a^2}{2}} \end{cases}$

$$\therefore \mathbb{E}(Z^2) = 1, \mathbb{E}(Z^4) = 3 \quad \begin{cases} M_Z'(a) = a e^{\frac{a^2}{2}}; M_Z''(a) = (1+a^2) e^{\frac{a^2}{2}} \\ M_Z'''(a) = (3a+a^3) e^{\frac{a^2}{2}}, M_Z^{(4)}(a) = (3+6a^2+a^4) e^{\frac{a^2}{2}} \end{cases}$$

$$\therefore \text{Var}(Z_t^2) = \mathbb{E}(Z_t^4) - (\mathbb{E}(Z_t^2))^2 = 3 - 1 = 2 \quad \therefore \text{Var}(Y) = \sum_{t=1}^n \text{Var}(Z_t^2) = 2n \quad \dots \text{as } Z_t \text{ i.i.d.}$$

Then, using central limit theorem (CLT).

as we can clearly see that X_t^2 has mean σ^2 and variance $2\sigma^4$ ($n=1$ case for $\text{Var}(\hat{\sigma}^2)$), then as each X_t^2 is i.i.d, the sample mean $\frac{1}{n} \sum_{t=1}^n X_t^2 = \hat{\sigma}^2$ ("sample") (and $\sigma^2 < \infty$)

$$\text{Satisfies } \underbrace{\sqrt{n}}_{\underbrace{\sum_{t=1}^n X_t^2}} (\underbrace{\hat{\sigma}^2 - \sigma^2}_{\underbrace{\mathbb{E}(X_t^2)}}) \xrightarrow{d} \underbrace{N(0, 2\sigma^4)}_{\underbrace{\text{Var}(X_t^2)}}$$

as above, we have that $\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n}$. Hence SE of $\hat{\sigma}^2$ is $\sqrt{\frac{2\sigma^4}{n}}$.

the 95% CI corresponding will be $\hat{\sigma}^2 \pm 1.96 \sqrt{\frac{2\hat{\sigma}^4}{n}}$.

Let $g(x) = \log(x)$. Hence, we have linear approximation for $\hat{\sigma}^2$ centered at σ^2 :

$$g(\hat{\sigma}^2) = g(\sigma^2) + g'(\sigma^2)(\hat{\sigma}^2 - \sigma^2) + \frac{1}{2} g''(\xi)(\hat{\sigma}^2 - \sigma^2)^2 \text{ for some } \xi \text{ between } \hat{\sigma}^2 \text{ and } \sigma^2.$$

$$\text{then, } \log(\hat{\sigma}^2) - \log(\sigma^2) = \frac{1}{\sigma^2} (\hat{\sigma}^2 - \sigma^2) - \frac{1}{2\xi^2} (\hat{\sigma}^2 - \sigma^2)^2$$

$$\text{multiplying by } \sqrt{n}, \text{ we have } \sqrt{n} (\log(\hat{\sigma}^2) - \log(\sigma^2)) = \underbrace{\frac{1}{\sigma^2} \sqrt{n} (\hat{\sigma}^2 - \sigma^2)}_{\text{main term}} - \underbrace{\frac{\sqrt{n}}{2\xi^2} (\hat{\sigma}^2 - \sigma^2)^2}_{\text{remainder } R_n}$$

$$\text{as we have shown } \sqrt{n} (\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$$

$$\text{the main term satisfies } \frac{1}{\sigma^2} \sqrt{n} (\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, \frac{2\sigma^4}{(\sigma^2)^2}) = N(0, 2)$$

for the remainder term R_n .

$$\text{Since } \sqrt{n} (\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4), \text{ we have } \sqrt{n} (\hat{\sigma}^2 - \sigma^2) = o_p(1)$$

$$\because \hat{\sigma}^2 - \sigma^2 = o_p(n^{-\frac{1}{2}}), (\hat{\sigma}^2 - \sigma^2)^2 = o_p(n^{-1}), \sqrt{n} (\hat{\sigma}^2 - \sigma^2)^2 = o_p(n^{-\frac{1}{2}}) = o_p(1) \cdot \frac{1}{\sqrt{n}}$$

$$\text{as } n \rightarrow \infty, \text{ we have } n^{-\frac{1}{2}} \rightarrow 0, o_p(n^{-\frac{1}{2}}) \rightarrow o_p(1)$$

$$\text{given } \xi \text{ lies between } \hat{\sigma}^2 \text{ and } \sigma^2, \text{ when } n \rightarrow \infty, \xi \xrightarrow{P} \sigma^2 \text{ as } \hat{\sigma}^2 \xrightarrow{P} \sigma^2.$$

$$\text{Hence } \frac{1}{2\xi^2} \xrightarrow{P} \frac{1}{2\sigma^4} = o_p(1). \because \frac{\sqrt{n}}{2\xi^2} (\hat{\sigma}^2 - \sigma^2)^2 = o_p(1) \cdot o_p(1) = o_p(1)$$

$$\because R_n \xrightarrow{P} 0. \text{ Hence, we have } \sqrt{n} (\log(\hat{\sigma}^2) - \log(\sigma^2)) \xrightarrow{d} N(0, 2)$$